

AN EMPIRICAL COMPARISON OF THE PERFORMANCE OF ALTERNATIVE OPTION PRICING MODELS

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This paper presents a comparison of alternative option pricing models based neither on jump-diffusion nor stochastic volatility data generating processes. We assume either a smooth volatility function of some previously defined explanatory variables or a model in which discrete-based observations can be employed to estimate both path-dependence volatility and the negative correlation between volatility and underlying returns. Moreover, we also allow for liquidity frictions to recognize that underlying markets may not be fully integrated. The simplest models tend to present a superior out-of sample performance and a better hedging ability, although the model with liquidity costs seems to display better in-sample behavior. However, none of the models seems to be able to capture the rapidly changing distribution of the underlying index return or the net buying pressure characterizing option markets.

Keywords: Option pricing, conditional volatility, hedging, liquidity, net buying pressure.

(JEL G12, G13, C14)

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1. Introduction

It is well known that subsequent to the crash of October 1987, implied volatilities of European index options have exhibited a pronounced smile or smirk with a slope that is higher the shorter the life of the option is. These empirical regularities have generally been interpreted as reflecting heavily left-skewed risk-neutral index distributions with excess kurtosis.

A double-jump stock price model as per Duffie, Pan, and Singleton (2000) is designed to capture these demanding regularities simultaneously. This model accommodates stochastic volatility, return-jumps, and volatility jumps. A key characteristic of the model is that it allows for a flexible correlation between return-jumps and volatility-jumps where upward volatility-jumps may provoke downward return-jumps. This is interesting because it may capture high kurtosis without imposing more negative skewness. Moreover, it nests the stochastic volatility model of Heston (1993), the stochastic volatility with return-jumps model of Bates (1996), the volatility-jump model of Duffie, Pan, and Singleton (2000), and the double-jump with independent arrival rate model of Duffie, Pan, and Singleton (2000) and Eraker, Johannes and Polson (2003). Finally, it is easily extended to include the random-intensity or state-dependent model of Bates (2000) and Pan (2002)¹.

Unfortunately, even the papers with both diffusive stochastic volatility and independent return and volatility jumps are not able to fully explain the smirkness and excess kurtosis found in the cross-section of index options². In other words, the parameter values necessary to

¹See also the alternative routes recently proposed by Huang and Wu (2004) and Santa-Clara and Yan (2004). Huang and Wu suggest a time-changed Lévy processes allowing an infinite number of jumps within any finite interval to be able to capture highly frequent discontinuous movements in the index return. By contrast, the typical compound Poisson jump model generates a finite number of jumps within a finite time interval. The authors show that this new specification is well suited to the behavior of short-term options, while long-term options are better priced allowing randomness in the arrival rate of jumps. Santa-Clara and Yan propose a linear-quadratic jump-diffusion model in which the jump intensity follows explicitly its own stochastic process to allow the jump intensity to have its own separate source of uncertainty. They employ this model to estimate the *ex-ante* equity premium.

²See the evidence reported by Bakshi, Cao and Chen (1997), Bates (2000), Chernov and Ghysels (2000), Anderson, Benzoni and Lund (2002), Eraker, Johannes and Polson (2003), Bakshi and Cao (2003) and Fiorentini, León and Rubio (2002) using Spanish data. In any event, it seems also to be the case that there are prices for volatility and jump risk. The above models are well posited for allowing estimation

match the smile in index options appear inconsistent with the time series properties of the stock index returns. On the other hand, Bakshi and Cao (2003), using individual option data, conclude that return-jumps are of a higher-order of importance than volatility-jumps, and that incorporating correlated volatility-jumps offers the further potential to reconcile option prices especially deep-out-of-the-money puts. It should be recognized that this flexibility has not been yet allowed for index options; however, individual risk-neutral return distributions are far-less negatively-skewed and more peaked than the index counterpart. For this reason, this does not seem to be a helpful extension when testing models with index option data. In other words, characterizing high kurtosis without strong negative skewness does not seem to be as crucial with index return data as with individual data³.

Interestingly, there is an alternative point of view on the issues involved to explain option pricing data anomalies known as the net buying pressure hypothesis. As recently pointed out by Bates (2003), Whaley (2003), and Bollen and Whaley (2004) a more promising avenue of research than developing more elaborate theoretical models is the study of the option market participants' supply and demand for different option series. The limited capitalization of the market makers implies a limited supply of options. On the other hand, dynamic replication, implicitly assumed by previous models, ensures that the supply curve for all options is a horizontal line. Independently of how large demand is for buying options, price and implied volatility are unaffected. However, it is clear that market makers are not willing to sell an unlimited number of contracts of a given option. The larger their positions, the larger the expected hedging costs are in both the bid-ask spreads of other options and in the availability of other series needed to hedge positions. Hence, the current literature debate seeks to distinguish between the stochastic process and the net buying pressure explanations of the available empirical evidence regarding option pricing⁴.

of these risk premia by using both the time series data on stock returns and the panel data on option prices. See the papers by Pan (2002) and Garcia, Ghysels and Renault (2004).

³See the evidence reported by Bakshi, Kapadia and Madan (2003).

⁴As an example, Bollen and Whaley (2004) find that changes in implied volatility of index option and stock option series are directly related to net buying pressure from public order flow.

Models in the Duffie, Pan, and Singleton (2000) family assume that volatility may be inferred when it is in fact impossible exactly to filter a volatility variable from discrete observations of the continuous-time data generating process of the underlying asset. Moreover, the estimation risk associated with the relevant parameters is clearly substantial, and the potential for overparametrization as we move toward more complicated models becomes a crucial issue in empirical evidence. Finally, stochastic models with jumps in both returns and volatility are difficult to estimate, and closed-form affine expressions become increasingly difficult to obtain.

This paper reports alternative and more easily applicable empirical evidence on option pricing. We argue that before we further investigate more sophisticated models we should perform a comprehensive analysis of time-varying discrete volatility. In particular, we ask whether the volatility function is a smooth function of some underlying variables and check the daily predictive power of the estimated coefficients. At the same time, we incorporate the potential impact of transaction costs by recognizing that the underlying asset market and the option market are not integrated. In this sense, our paper adds new evidence associated with the current debate by noting that net buying pressure is related to the impact of transaction costs. Thus, discrepancies between the properties associated with prices in the two markets may be explained by liquidity costs that are idiosyncratic to the options market, and not to the underlying distribution process.

Along these lines, we compare five option pricing models, avoiding the stochastic framework but allowing for the potential impact of skewness, excess kurtosis, time-varying volatility and liquidity frictions⁵. The models are the traditional Black-Scholes (1973) method (BS hereafter), an *ad-hoc* BS method where the implied volatility is assumed to be a (parametric) quadratic function of the exercise price as suggested by Dumas, Fleming and Whaley (1998), a similar semiparametric model also extended by the explicit recognition of liquidity costs, and the Heston and Nandi (2000) (HN hereafter) GARCH option pricing model where volatility is readily predictable from the history of the underlying asset prices⁶. The comparison is made in terms of daily in-

⁵See Garcia, Ghysels and Renault (2004) for an excellent presentation of option pricing under a discrete time setting using the stochastic discount factor paradigm.

⁶As far as we know, this is the first evidence available on the HN model besides the results provided by the authors for the US market.

sample and out-of-sample pricing performance and hedging behavior of the models. To the best of our knowledge this is the first empirical evidence available that simultaneously analyzes, at least indirectly, the two current approaches to empirical option pricing.

Generally speaking, our out-of-sample empirical results tend to support the simplest models since the *ad-hoc* BS, and the univariate semi-parametric models lead to the smallest errors in terms of both pricing and hedging. On the hand, liquidity costs are clearly relevant when pricing is analyzed in-sample.

The rest of the paper is organized as follows. Section 2 contains a brief discussion of the five competing models used in the research and Section 3 describes the option data employed in the paper. The results regarding the pricing performance of all models are shown in Section 4, while Section 5 contains the hedging evidence. Section 6 analyzes the structure of pricing errors, while Section 7 provides some final remarks and concludes. Technical details on several of the models are relegated to Appendices A.1 and A.2.

2. Competing option pricing models and estimation details

2.1 The traditional Black-Scholes method

Under the BS assumptions, the well known option pricing formula for European calls on futures of Black (1976) is given by

$$c = e^{-r\tau} FN(d_1) - XN(d_2) \quad [1]$$

where

$$d_1 = \frac{\ln(F/X) + (\sigma^2/2)\tau}{\sigma\sqrt{\tau}}$$

and

$$d_2 = d_1 - \sigma\sqrt{\tau}$$

As usual, c denotes the price of a given call option with exercise price X and time to maturity τ . F is the future price of the underlying asset, r the riskless asset return and σ the volatility in the diffusion of the underlying asset. In a pure BS framework, the volatility is constant and the only parameter that must be estimated for option pricing. In the empirical application the implied volatility for each date t is obtained by minimizing the sum of squared pricing errors over all options available on that particular day. This volatility is

employed to price all options over the next date $t + 1$ when our main out-of-sample tests are run, and the contemporaneous day t when in-sample analysis is performed.

2.2 The ad-hoc Black-Scholes method

This is the model inspired by the deterministic framework suggested by Dumas, Fleming and Whaley (1998), in which each option has its own implied volatility depending on the exercise price (and time to maturity when applicable). Thus, the spot volatility of the underlying asset is a parametric quadratic function of the exercise price X ,

$$\hat{\sigma}(X) = a_0 + a_1X + a_2X^2 \quad [2]$$

where the coefficients are estimated every day by OLS, minimizing the sum of squared errors between the BS implied volatilities across different exercise prices and the model functional form of the implied volatility. When testing out-of-sample, these estimates are employed to obtain the volatility for each exercise price at day $t + 1$, whereas day t is used for the in-sample analysis. In the out-of-sample context, this has been shown to be a very useful approach since, in fact, it prices options with the smile observed on the previous day. It turns out that the idea may be rationalized by recalling the expression proposed by Backus, Foresi, Li and Wu (1997) who adopt a Gram-Charlier series expansion of the normal density function to obtain skewness and kurtosis adjustment terms for the BS formula on the basis of the insight due to Jarrow and Rudd (1982). It can be shown that,

$$\sigma(F/X) \cong \sigma\sqrt{\tau} \left[1 - \frac{d_1}{3!}sk + \frac{d_1^2 - 1}{4!}ku \right] \quad [3]$$

where $\sigma(F/X)$ is the implied volatility smile, σ is the (annualized) BS volatility, sk is the unconditional skewness under the risk neutral measure of returns, and ku is the corresponding unconditional excess kurtosis⁷. Expression [3] can be written as

$$\sigma(F/X) \cong \beta_0 + \beta_1d_1 + \beta_2d_1^2 \quad [4]$$

⁷We write the volatility function given by [3] in the traditional way as a function of the moneyness degree, while expression [2] is written in terms of the exercise price. It should be noted that both expressions reflect exactly the same idea, given that the level of the future price for equation [3] to make sense must remain constant.

where β_0, β_1 and β_2 are

$$\begin{aligned}\beta_0 &\equiv \sigma\sqrt{\tau} - \frac{\sigma\sqrt{\tau}}{4!}ku \\ \beta_1 &\equiv -\frac{\sigma\sqrt{\tau}}{3!}sk \\ \beta_2 &\equiv \frac{\sigma\sqrt{\tau}}{4!}ku\end{aligned}$$

and therefore the quadratic effects of the exercise price of equation [2] may be associated with the underlying skewness and kurtosis of the underlying asset distribution function. This is probably the simplest way, although not used previously in literature, to relate smile patterns with the distribution of the underlying asset.

Dumas, Fleming and Whaley (1998) also consider time to maturity as an explanatory variable. In our available dataset, time to maturity is not a relevant variable since, for each fixed day, all options used trade for the nearest expiration date. Therefore, in the empirical application only the exercise price appears in this *ad-hoc* BS version of the model.

2.3 A univariate nonparametric method

Here a more flexible estimation of the implied volatility pattern is allowed. Instead of estimating a quadratic or any other parametric specification of volatility in terms of exercise price, a nonparametric estimator will be used that only assumes volatility to be a smooth function of exercise price. A similar approach is considered by Aït-Sahalia and Lo (1998). However, we propose a different nonparametric technique that, as we argue below, is especially appropriate for option pricing data. In particular, the Symmetrized Nearest Neighbor (SNN) estimator is selected, since it presents better properties when the explanatory variable does not present a uniform design. In our case, the explanatory variable is the exercise price, which is far from being uniform, and the SNN estimation procedure rather than the traditional kernel estimator is the natural choice. This point has not been recognized by previous empirical applications of nonparametric methods to option data.

Let us briefly describe both kernel and SNN nonparametric regressions. Given the options available on a particular day t , the dataset Ω_t will contain the information $\{\sigma_i, X_i\}_{i \in t}$ where σ_i denotes the implied volatility for the i th option, and X_i its exercise price. Consider

now an option to be priced at $t + 1$ with exercise price X . Given the information set Ω_t , the kernel estimator for the implied volatility is

$$\hat{\sigma}_K(X) = \frac{\frac{1}{nh} \sum_{i \in t} K\left(\frac{X - X_i}{h}\right) \sigma_i}{\frac{1}{nh} \sum_{i \in t} K\left(\frac{X - X_i}{h}\right)} \quad [5]$$

where K denotes a second order kernel (a function that basically behaves as a density) that assigns the weight to each value in the dataset. The value h is the bandwidth or smoothing parameter that, in general, is selected using a data-driven method. If h is too small, the estimator picks up the cyclical nature of the data but also random variations due to noise. In this case, the bias is reduced whereas the variance increases. By contrast, if h is too large, the genuine variation of the function along with the noise is eliminated, and the variance is reduced but the bias increases. A proper selection of h seeks to minimize the mean square error (MSE), providing a compromise between bias and variance. Looking at the expression for the MSE, given in Appendix A.1, it can be seen that the optimal value for h should depend on the density of the explanatory variable. In fact, a higher value of h will be desirable for those zones where the density is low and a lower value where the density is high. To deal with this fact a kernel with a variable smoothing parameter might be a solution, although it turns out to be difficult to implement and its performance in practice is not satisfactory⁸.

Alternatively, an SNN estimator that plays the role of a variable kernel presents theoretical and practical advantages. This estimator was proposed by Yang (1981) and studied in detail by Stute (1984). When estimating at one point (X_i) we calculate the weight for the rest of observations by looking at the distance between the values of the empirical distribution at each point ($F_n(X_i)$) rather than the distance between the points themselves. Hence, the estimator is defined as

$$\hat{\sigma}_{SNN}(X) = \frac{\frac{1}{nh} \sum_{i \in t} K\left(\frac{F_n(X) - F_n(X_i)}{h}\right) \sigma_i}{\frac{1}{nh} \sum_{i \in t} K\left(\frac{F_n(X) - F_n(X_i)}{h}\right)} \quad [6]$$

⁸See Härdle (1990) for a general review of nonparametric methods.

where $F_n(\cdot)$ denotes the empirical distribution of the exercise price X . It turns out that, as opposed to the usual kernel estimator, the variance in the MSE does not depend on density and, under very general conditions, if an SNN estimator is employed, the MSE is smaller on the tails of the design. This is illustrated in Appendix A.1. Typically, these are the most interesting zones when working with option data since they correspond to in-the-money and out-of-the-money options. In addition, the SNN method is very easy to compute since the expression can be computed as a usual kernel estimator once the empirical distribution is obtained.

2.4 A bivariate nonparametric method

Longstaff (1995), Peña, Rubio and Serna (1999) and Ferreira, Gago and Rubio (2003) report evidence relating the smile and the liquidity costs as proxied by the bid-ask spread. Motivated by these results, the bid-ask spread (SP) is added as another explanatory variable in a nonparametric framework. To remain consistent with the univariate estimator described above, we use the extension of the univariate SNN estimator to the bivariate case. The bivariate estimator is then defined as

$$\hat{\sigma}_{SNN}(X) = \frac{\frac{1}{nh_X h_{SP}} \sum_{i \in t} K\left(\frac{F_n(X) - F_n(X_i)}{h_X}\right) K\left(\frac{F_n(SP) - F_n(SP_i)}{h_{SP}}\right) \sigma_i}{\frac{1}{nh_X h_{SP}} \sum_{i \in t} K\left(\frac{F_n(X) - F_n(X_i)}{h_X}\right) K\left(\frac{F_n(SP) - F_n(SP_i)}{h_{SP}}\right)} \quad [7]$$

where the information set for each day will also include the bid-ask spread of the options; that is, $\Omega_t = \{\sigma_i, X_i, SP_i\}_{i \in t}$. The statistical properties from the univariate estimator do not translate straightforwardly to the multivariate setting. However, the main result remains true and, under very general conditions, the performance of the bivariate SNN estimator is better in the tails of the marginal distributions for the covariates. Some theoretical results are also provided in Appendix A.1 and a more detailed study can be found in Ferreira and Gago (2002), where a simulation study is carried out that supports the theoretical results.

In the implementation of our two nonparametric regressions discussed above, a Gaussian kernel has been selected, although any other second order kernel would lead to similar results. Several methods for selecting smoothing parameters can be considered for the univariate case.

As a general classification, we have the so called plug-in approach, leave-one-out methods and methods based on penalizing the sum of residuals⁹. The resulting estimator depends crucially on the parameter choice and therefore, the selection procedures matters. Since different methods may end up with different estimations, we have computed the estimators using three different alternatives. The first is the univariate plug-in method of Gasser, Kneip and Köhler (1991). The second and third are the natural extensions of the Generalized Cross Validation (GCV) and Rice criteria to the bivariate case. As it turns out, the results are very similar for all of them and the main conclusions remain the same independently of the method used in practice. In the empirical results below we report only the estimations obtained with the Rice method.

Let us describe the steps involved in the empirical application. For each option traded on day t , the procedure to estimate the implied volatility goes as follows: 1) the smoothing parameters are computed each month with all data available and remain fixed until the next month; 2) the bid-ask spread is estimated as the average of the relative bid-ask of all options of the same class, with the same time to maturity and exercise price, traded on t and $t + 1$ until just before the corresponding option is observed; 3) given the smoothing parameters and the bid-ask spread estimate, the implied volatility is estimated using the information set Ω_t and applying the formula for $\hat{\sigma}_{SNN}(X)$ in the univariate case and $\hat{\sigma}_{SNN}(X, SP)$ in the bivariate case; 4) the estimated volatility is then plugged in the BS formula to price all options at $t + 1$ ¹⁰. This implies that under the SNN nonparametric volatility functions both with and without liquidity costs we obtain semiparametric theoretical option prices¹¹.

Some remarks are in order. The univariate nonparametric method is similar in spirit to the *ad-hoc* volatility estimation procedure. The difference relies on the flexibility of the implied volatility function allowed by the SNN procedure. The price for flexibility is a slower rate of convergence to the true function. This means that if the function is indeed quadratic, as considered in the *ad-hoc* BS model, this method should better capture this pattern. However, if the function presents a

⁹See Härdle (1990).

¹⁰In the in-sample analysis only information up to date t is employed.

¹¹This is like Art-Sahalia and Lo (1998). See their discussion on the dimensionality problem.

different pattern, the *ad-hoc* BS method leads to wrong estimates while the nonparametric estimator is still consistent, due to its independence of the particular specification for the volatility function.

2.5 The HN GARCH method

All the above methods are useful for practical purposes although they do not explicitly explain the reasons for the underlying path that determines the volatility pattern; they just take this pattern as given and try to estimate it by the most reasonable procedure. They are based merely on the simple idea that an adaptive estimation of the implied volatility should improve the BS pricing formula.

As a final competing model, we consider the HN (2000) GARCH(1,1) option pricing formula as particularly appealing because it permits us to capture the path dependence in volatility as well as the negative correlation of volatility with index returns. Moreover, a closed-form pricing formula is obtained. From a practical point of view, the HN model presents an advantage with respect to Heston's (1993) continuous-time approach. The model is discrete and the parameters can be estimated using maximum likelihood. Moreover, the asymmetric GARCH(1,1) version has a continuous limit that is identified as Heston's model with perfect correlation between the underlying asset and the volatility. This implies that neither the volatility risk premium nor the jump-risk premium exist in this framework. Also, for the purpose of comparing pricing methods, the HN model can be considered as a model with predictable volatility, since volatility can be estimated from the past information of the underlying return path. Therefore, a good empirical performance would provide evidence in favour of deterministic volatility models, frequently updated, rather than models considering new risk factors. Once again, the idea is to check whether this simple discrete-time model is rich enough to incorporate the peculiarities of option prices.

The HN model considers that the log-spot asset price, $\ln S(t)$, is characterized by the following GARCH(1,1) process:

$$\ln S(t) = \ln S(t - \Delta) + r + \lambda h(t) + \sqrt{h(t)}z(t) \quad [8]$$

$$h(t) = \omega + \beta h(t - \Delta) + \alpha \left(z(t - \Delta) - \gamma \sqrt{h(t - \Delta)} \right)^2 \quad [9]$$

where $h(t)$ is the conditional variance of the log return between $t - \Delta$ and t and is predictable from the information set at time $t - \Delta$; λ

is the risk premium embedded in returns; $z(t)$ is a standard normal disturbance, and the parameter γ controls the skewness of the distribution.

The model allows for the prediction of $h(t + \Delta)$ at time t from the spot price as

$$h(t + \Delta) = \omega + \beta h(t) + \alpha \frac{(\ln S(t) - \ln S(t - \Delta) - r - \lambda h(t) - \gamma h(t))^2}{h(t)} \quad [10]$$

where α determines the kurtosis of the distribution, and because of the asymmetric parameter, γ , a large negative shock, $z(t)$, raises the variance more than a large positive shock. This model enables us to capture the observed negative correlation between returns and variance. In particular,

$$\text{cov}_{t-\Delta}[h(t + \Delta), \ln S(t)] = -2\alpha\gamma h(t) \quad [11]$$

where, given a positive α , positive values for γ result in negative correlation between returns and variance.

To obtain the pricing formula, the process is written in terms of the risk-neutral measure and it can be shown that the value of a call option is given by

$$c = e^{-r\tau} [F(t) P_1 - X P_2] \quad [12]$$

where, as in the BS formula, P_2 is the risk neutral probability of the asset being greater than X at maturity and P_1 corresponds to the delta of the call value. Of course, the computation of P_1 and P_2 in this case requires estimates of the risk-neutral parameters in the GARCH model. A more detailed description of the pricing procedure is given in Appendix A.2.

For empirical analysis the model is estimated daily and for the out-of-sample tests the parameters obtained with data up to day t are used to price the options traded at $t + 1$. As with other models, for the in-sample case, only information available up to date t is employed to price options traded at t . Thus, the parameters are estimated daily and recursively with two years and one month of past data, or 522 daily observations of the index return¹². The only parameter that

¹²We have both closing price daily returns for the IBEX-35 from January 1994 to November 1998, and 15-minute returns from 1996 to 1998. The daily returns are used in the estimation of the NH GARCH model, and it is therefore the basic

needs to be changed in the option pricing formula is the skewness parameter γ^* which, under the risk neutral process, turns out to be equal to $\gamma^* = \gamma + \lambda + 1/2$. Proceeding in this way, the model does not really need data from option prices. However, in order to incorporate simultaneously information from the option and common stock markets, the risk neutral skewness parameter γ^* is implicitly computed from the cross-section of option prices. The implicit estimator is defined as the minimizer of the pricing errors in the options traded at time t ; that is, $\gamma_t^* = \arg \min_{\{\gamma\}} \sum_{i \in t} (c_{HN,i}(\gamma) - c_i)^2$, where c_i is the price observed in the market, and $c_{HN,i}(\gamma)$ denotes the price resulting from the HN formula, where the rest of parameters are those estimated from the underlying asset return time series. Given all parameter estimates available at data t , the HN formula is applied to obtain the out-of-sample price of each option at $t + 1$, and the in-sample price at day t .

As discussed above, and unlike the continuous-time stochastic volatility version of the model, all the parameters may be estimated directly from the daily closing returns of the IBEX-35 index using the well known maximum likelihood estimation proposed by Bollerslev (1986). To illustrate the behavior of the GARCH model, Table 1 shows the maximum likelihood estimates of the model on the daily IBEX-35 data from January 1996 to November 1998. The skewness parameter, γ , is positive and significantly different from zero, indicating that shocks to returns and volatility covary negatively. The parameter that measures the degree of mean reversion, given by $\beta + \alpha\gamma^2$ and equal to 0.944, is higher than the estimate reported by HN for the US market. The volatility of volatility, α , is small but also statistically significant, and the annualized long-run mean of volatility, as given by $\sqrt{252(\omega + \alpha) / (1 - \beta - \alpha\gamma^2)}$ (assuming 252 trading days), is 21.7%. Although not reported, the restricted version of the model with $\gamma = 0$ is rejected using both the log-likelihood ratio test and the SIC statistic.

data to estimate the conditional daily volatility during our sample period. The rationality of using two years of daily past data to estimate parameters comes from the experience of León and Mora (1999) when estimating several models of the GARCH family with Spanish data. The additional month is due to the maximum length period of the nearest expiration contracts of the options in our dataset.

TABLE 1
The Heston-Nandi GARCH estimators

The parameters are estimated with daily spot index returns from February 1, 1996 to November 10, 1998 using the following model^{1/}:

$$\ln S(t) = \ln S(t-1) + r + \lambda h(t) + \sqrt{h(t)}z(t)$$

$$h(t) = \omega + \beta h(t-1) + \alpha \left(z(t-1) - \gamma \sqrt{h(t-1)} \right)^2$$

where $h(t)$ is the conditional variance of the log return between $t-1$ and t and is predictable from the information set at time $t-1$; $z(t)$ is a standard normal disturbance, and the parameter γ controls the skewness of the distribution.

α	β	γ	ω	λ
0.0000104 (1.877)	0.9226 (29.701)	45.718 (2.159)	1.00e ⁻²⁰ (0.000)	4.264 (1.339)

1/ The long term variance is $\theta = \sqrt{252(\omega + \alpha) / (1 - \beta - \alpha\gamma^2)} = 0.2170$, and the persistence parameter is $\beta + \alpha\gamma^2 = 0.944$.

3. Option data description

The Spanish IBEX-35 index is a value-weighted index comprising the 35 most liquid Spanish stocks traded in the continuous auction market system. The official derivative market for risky assets, which is known as MEFF, trades a futures contract on the IBEX-35, the corresponding option on the IBEX-35 futures contracts for calls and puts, and individual option contracts for blue-chip stocks.

The Spanish option contract on the IBEX-35 futures is a cash settled European option with trading over the three nearest consecutive months and the other three months of the March-June-September-December cycle. The expiration day is the third Friday of the contract month. Prices are quoted in full points, with a minimum price change of one index point. The exercise prices are given by 50 index point intervals.

Our database comprises of all call and put options on the IBEX-35 index futures traded daily on MEFF during the period January 1996 through November 1998¹³. Liquidity is concentrated on the nearest expiration contract. In fact, during the sample period almost 90% of trades occurred in this type of contract. Given the concentration in

¹³It may seem appropriate to extend the sample period. However, the structural organization of the Spanish market and its liquidity has changed importantly over time. We argue that our sample period may be characterized as a very homogeneous time period and, in this sense, it is quite useful for our objectives.

liquidity, our daily set of observations includes only calls and puts with the nearest expiration date.

For each option traded we have the transaction price, the relative bid-ask spread, the exercise price, the expiration date, the simultaneous future price as measured by its bid-ask spread average, and the annualized repo T-bill rates with approximately the same maturity as the option.

We restrict our attention to options transacted from 11:00 to 16:45. Every trade recorded during this window is used in the estimation. Note that care has also been taken to eliminate the artificial trading potential problems associated with market makers margin requirements, and also with the well known intraday seasonalities on the underlying index behavior. Finally, we eliminate all call and put prices that violate the well known arbitrage bounds given by¹⁴,

$$\begin{aligned} c &\geq (F - X)e^{-r\tau} \\ p &\geq (X - F)e^{-r\tau} \end{aligned}$$

These exclusionary criteria yield a final daily sample of 30542 observations (18656 calls and 11886 puts). Table 2 describes the sample properties of the call and put option prices employed in this work. Average prices, average relative bid-ask spread, average number of contracts per day and the number of options are reported for each moneyness category. Moneyness is defined as the ratio of the exercise price to futures price. A call (put) option is said to be deep out-of-the-money (deep in-the-money) if the ratio X/F is greater than 1.03; out-of-the-money (in-the-money) if $1.03 \geq X/F > 1.01$; at-the-money when $1.01 \geq X/F > 0.99$; in-the-money (out-of-the-money) when $0.99 \geq X/F > 0.97$; and deep-in-the-money (deep out-of-the-money) if $0.97 > X/F$. The average option price ranges from 60.7 pesetas for deep out-of-the-money calls (deep in-the-money puts) to 110.2 pesetas for at-the-money options. As expected, the extreme options (in terms of moneyness) have the highest bid-ask spreads. In other words, deep out-of-the-money (in-the-money) options have the highest liquidity cost, while at-the-money options have the lowest.

¹⁴ Approximately 1.1% of all options in the dataset violated these arbitrage bounds.

TABLE 2
Sample characteristics of IBEX-35 futures options

Options	Moneyness	Average Price	Average Relative Bid-Ask Spread	Average Number of Contracts	Number of Options
DOTMC (DITMP)	>1.03	60.74	0.3173	45.7	5198
OTMC (ITMP)	1.01-1.03	82.57	0.1969	75.9	6613
ATMC (ATMP)	0.99-1.01	110.18	0.1343	76.9	9157
ITMC (OTMP)	0.97-0.99	100.78	0.1606	89.5	5218
DITMC (DOTMP)	<0.97	78.82	0.2563	66.0	4356
ALL OPTIONS	-	89.71	0.2009	71.9	30542

Average prices, average relative bid-ask spread and the average number of contracts per day are reported for each moneyness category. All call and put options transacted over the interval from 11:00 to 16:45 are employed from January 2, 1996 to November 10, 1998. Moneyness is defined as the ratio of the exercise price to the futures prices. DOTM, OTM, ATM, ITM and DITM are deep-out-of-the money, out-of-the-money, at-the-money, in-the-money, and deep-in-the-money options respectively. C or P indicates call or put.

4. Pricing performance

4.1 Testing the statistical performance of competing models

This section reports both in-sample and out-of-sample daily pricing performance of the five competing models analyzed by our paper. The statistical significance of performance for in-sample and out-of-sample pricing errors is assessed first by analyzing the proportions of theoretical prices lying outside their corresponding bid-ask boundaries. Then we test whether or not the differences between proportions of any two competing models are statistically different from zero, taking into account that any two competing models are not independent.

Specifically, we take any pair of two models. Let p^1 be the proportion of calls (puts) whose theoretical price lies outside the bid-ask spread when we price with model 1, and let p^2 be the corresponding proportion when we price with model 2. Also, let Z^1 be 1 if the theoretical price (for model 1) is outside the spread and 0 otherwise. Finally, Z^2 is 1 if the theoretical price (for model 2) is outside the spread and 0 otherwise. Then, $Z^1 - Z^2$ equals -1 with probability π_1 , 0 with probability π_2 , and 1 with probability $1 - \pi_1 - \pi_2$. Under the null hypothesis of equal proportions,

$$\begin{aligned} E(Z^1 - Z^2) &= 0 & \implies \pi_2 &= 1 - 2\pi_1 \\ \text{var}(Z^1 - Z^2) &= 2\pi_1 \end{aligned}$$

Note that,

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n Z_i^1 &= p^1 \\ \frac{1}{n} \sum_{i=1}^n Z_i^2 &= p^2\end{aligned}$$

Consider the statistic defined as,

$$Z = \frac{1}{n} [(Z_1^1 - Z_1^2) + \dots + (Z_n^1 - Z_n^2)] \quad [13]$$

by the Central Limit Theorem,

$$Z \rightarrow N\left(0, \frac{2\pi_1}{n}\right)$$

where π_1 can be estimated as

$$\hat{\pi}_1 = \frac{\text{no.of ones} + \text{no.of minus ones}}{2n}$$

Since the differences in proportions coincide with the Z-statistic, then the final statistic employed to compare the models is given by

$$Z = p^1 - p^2 \rightarrow N\left(0, \frac{2\hat{\pi}_1}{n}\right) \quad [14]$$

We also compare the performance of all models by analyzing prices inferred from each theoretical model against the observed market prices. In particular, we compute for each model the absolute pricing error as given by the square root of the squared differences between the theoretical price and the market price of each option i :

$$APE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{i,model} - P_{i,mket})^2} \quad [15]$$

where n is the number of options, and $P_{i,model}$ and $P_{i,mket}$ are the theoretical price for either a call or a put for each of the five models, and the observed market price respectively.

To test statistically whether the average absolute pricing errors of two competing models are significantly different from zero, we perform a GMM overidentifying restriction test, with the Newey-West weighting

covariance matrix. We consider the following set of moment conditions for any two competing models

$$\begin{pmatrix} n^{-1} \sum_{i=1}^n |APE_i^1| - m \\ n^{-1} \sum_{i=1}^n |APE_i^2| - m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad [16]$$

where $APE_i^{1,2}$ is the absolute pricing error for option i and either model 1 or 2, and m is the common mean pricing error under the null hypothesis that both models have the same pricing error. The test statistic is distributed as a χ^2 with one degree of freedom.

4.2 In-sample pricing performance

The statistical in-sample performance of all models using the proportion that the theoretical price is outside the bid-ask spread is contained in Table 3 for both calls and puts. The figures in each panel are the proportions of theoretical prices lying outside the spread for options within a moneyness category¹⁵, and below the identifying number of the competing model that has a significantly lower proportion than the model being analyzed according to our Z-statistic in equation [14]. As an example, and in order to facilitate the reading of the following tables, note that the semiparametric model with liquidity costs has 16.4% of all call prices outside the observed bid-ask spread. Moreover, analyzing the statistical significance of that proportion relative to the proportion associated with any other model by Z-statistic, we conclude that this proportion is statistically different (lower) than the proportions given by BS (model 1), the semiparametric model with the nonparametric volatility function depending only on the exercise price (model 2), the *ad-hoc* BS (model 4), and HN (model 5). Therefore, for call options the flexible semiparametric model with liquidity costs presents a statistically better performance than the rest of models. Interestingly, however, ATM calls are better priced in the sense defined above by the *ad-hoc* BS. As expected, under the net buying pressure hypothesis and for calls, liquidity costs are more relevant for options with extreme moneyness degrees. At the same time, theoretical put prices always present statistically lower proportions outside the

¹⁵ In Tables 3 to 7, deep in-the-money (out-of-the-money) calls (puts), and deep out-of-the-money (in-the-money) calls (puts), as reported in Table 1, are now included in the corresponding out-of-the-money and in-the-money categories.

bid-ask boundary when liquidity costs are included. The flexibility of the nonparametric estimation of the volatility function together with liquidity costs seems to be a key consideration when pricing options in-sample. It is important to note, however, that the key in understanding the results lies in the impact of liquidity costs rather than in the flexible functional form of the volatility function. The *ad-hoc* BS has lower proportions of theoretical prices outside the spread when compared to the semiparametric model with the exercise price as the only explanatory variable. Indeed, the quadratic parametric representation of the volatility function seems to be, on average, a good representation of the volatility smile.

TABLE 3
Daily in-sample pricing performance of alternative option pricing models for calls and puts

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	0.470 Z (5)	0.205 Z (1,5)	0.164 Z (1,2,4,5)	0.180 Z (1,2,5)	0.579 Z (-)
OTM	0.455 Z (5)	0.211 Z (1,5)	0.160 Z (1,2,4,5)	0.190 Z (1,2,5)	0.631 Z (-)
ATM	0.456 Z (5)	0.195 Z (1,5)	0.180 Z (1,2,5)	0.169 Z (1,2,3,5)	0.528 Z (-)
ITM	0.470 Z (-)	0.200 Z (1,5)	0.137 Z (1,2,4,5)	0.157 Z (1,2,5)	0.428 Z (1)
1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	0.471 Z (5)	0.252 Z (1,5)	0.181 Z (1,2,4,5)	0.200 Z (1,2,5)	0.555 Z (-)
OTM	0.499 Z (5)	0.271 Z (1,5)	0.186 Z (1,2,4,5)	0.208 Z (1,2,5)	0.541 Z (-)
ATM	0.394 Z (5)	0.217 Z (1,5)	0.164 Z (1,2,4,5)	0.190 Z (1,2,5)	0.579 Z (-)
ITM	0.517 Z (5)	0.213 Z (1,5)	0.158 Z (1,2,4,5)	0.197 Z (1,5)	0.595 Z (-)

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the contemporaneous (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; *Ad-hoc* BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. The statistical performance for the pricing errors is assessed by analyzing the proportion of theoretical prices lying outside their bid-ask spread boundaries; that is, $\text{Prob}(\text{Price}_{\text{model}} \notin [\text{Bid}, \text{Ask}])$. In each case we report the corresponding proportion for a given model and, in parentheses, the identifying number of the competing model that has a significantly higher proportion than the model being analyzed according to our Z-statistics at the 5% level. OTM, ATM, ITM are out-of-the-money, at-the-money, in-the-money options respectively. X is the exercise price, and SP is the relative bid-ask spread.

When performance is analyzed in terms of pricing differences by the *APE* statistic given by equation [15] and the χ^2 GMM test in [16], we find very similar results. The results are reported in Table 4. Again, liquidity costs seem to be a key consideration when pricing options in-sample. Note that even ATM calls have less pricing differences. Given that these options have lower proportions outside the spread when pricing by *ad-hoc* BS, we may conclude that options which belong to the 16.9% outside the boundaries, as reported in Table 3, must have quite large pricing differences to justify the worse performance found under the *APE* statistic in Table 4.

TABLE 4
Daily in-sample absolute mean pricing errors of alternative option pricing models for calls and puts

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	6.417 χ^2 (5)	3.016 χ^2 (1,5)	1.677 χ^2 (1,2,4,5)	2.843 χ^2 (1,2,5)	8.422 χ^2 (-)
OTM	5.585 χ^2 (5)	2.793 χ^2 (1,5)	1.639 χ^2 (1,2,4,5)	2.690 χ^2 (1,2,5)	9.144 χ^2 (-)
ATM	6.423 χ^2 (5)	3.253 χ^2 (1,5)	1.839 χ^2 (1,2,4,5)	3.037 χ^2 (1,2,5)	7.469 χ^2 (-)
ITM	6.749 χ^2 (5)	3.590 χ^2 (1,5)	1.390 χ^2 (1,2,4,5)	3.136 χ^2 (1,2,5)	7.152 χ^2 (-)
1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	5.531 χ^2 (5)	3.504 χ^2 (1,5)	2.124 χ^2 (1,2,4,5)	2.684 χ^2 (1,2,5)	6.975 χ^2 (-)
OTM	5.501 χ^2 (5)	3.724 χ^2 (1,5)	2.439 χ^2 (1,2,5)	2.558 χ^2 (1,2,5)	6.488 χ^2 (-)
ATM	4.903 χ^2 (5)	2.894 χ^2 (1,5)	1.508 χ^2 (1,2,4,5)	2.675 χ^2 (1,2,5)	7.581 χ^2 (-)
ITM	7.325 χ^2 (5)	3.870 χ^2 (1,5)	1.606 χ^2 (1,2,4,5)	3.916 χ^2 (1,5)	8.154 χ^2 (-)

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the contemporaneous (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; *Ad-hoc* BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. The pricing error is computed as the square root of the average of the squared difference between the theoretical price and the market price for each model:

$$APE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{i,model} - P_{i,market})^2}$$

The statistical performance for the pricing errors is assessed by a GMM overidentifying test with the Newey-West weighting covariance matrix. The test statistic is distributed as a χ^2 with one degree of freedom. In each case we report the corresponding absolute value pricing error for a given model and, in parentheses, the identifying number of the competing model that has a significantly different (higher) pricing error than the model being analyzed according to our χ^2 -statistics at the 5% level. OTM, ATM, ITM are out-of-the-money, at-the-money, in-the-money options respectively. X is the exercise price, and SP is the relative bid-ask spread.

Finally, both the traditional BS model and the HN GARCH(1,1) present a very poor performance both in terms of proportions outside the bid-ask spread and in terms of pricing differences. We will come back to these results when analyzing out-of-sample performance.

4.3 Out-of-sample pricing performance

Of course, from the practitioner's point of view, it is more relevant to know how option pricing models value derivatives out-of-sample. This has always been the important perspective in option modeling. The results using proportions off the spread are contained in Table 5. As before, they are the frequencies of theoretical prices lying outside the bid-ask spread, and below the identifying number of the competing model that has a significantly lower proportion than the model being analyzed according to our Z-statistic. In this case, and contrary to the in-sample evidence, the *ad-hoc* BS dominates all models with 41.0% (39.6%) of all call (put) prices outside the observed bid-ask spread. Moreover, the proportions observed in the *ad-hoc* BS case are statistically lower for all moneyness categories except for in-the-money options. Therefore, the *ad-hoc* BS presents a statistically superior out-of-sample performance than the rest of option pricing models considered in the paper¹⁶.

Surprisingly, the out-of-sample results show that the univariate non-parametric pricing method obtains significantly lower proportions outside the spread than the model with liquidity costs. First, the fact that a parametric model outperforms the more flexible approach indicates that again, on average, the symmetric smile rather than the downward sloping smirk captures the behavior of the distribution of the underlying asset during this sample period. Note that this *ad-hoc* parametric model is consistent with pricing by market makers in actual trading. Secondly, the impact of moving from in-sample to out-of-sample is quite dramatic. The proportions outside the spread increase by 100%

¹⁶Using quarterly data, as in Ferreira, Gago and Rubio (2003), the results employing the exercise price or the moneyness degree, as measured by X/F , in the estimation of the volatility function are the same. Note that when using moneyness the issue is to decide the appropriate underlying price. In principle, it should be the same for different exercise prices. In the application with quarterly data, the average of the underlying price during each sample period is taken as a proxy for the denominator. However, to employ this ratio with daily data is more problematic and it is discarded from the estimation.

TABLE 5
Daily in-sample pricing performance of alternative option pricing
models for calls and puts

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	0.516 Z (5)	0.418 Z (1,3,5)	0.447 Z (1,5)	0.410 Z (1,2,3,5)	0.657 Z (-)
OTM	0.579 Z (5)	0.461 Z (1,3,5)	0.479 Z (1,5)	0.456 Z (1,2*,3,5)	0.709 Z (-)
ATM	0.419 Z (5)	0.375 Z (1,3,5)	0.406 Z (1,5)	0.356 Z (1,2,3,5)	0.618 Z (-)
ITM	0.446 Z (5*)	0.302 Z (1,3,5)	0.383 Z (1,5)	0.304 Z (1,3,5)	0.469 Z (-)
1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	0.565 Z (5)	0.424 Z (1,3,5)	0.504 Z (1,5)	0.396 Z (1,2,3,5)	0.614 Z (-)
OTM	0.588 Z (-)	0.446 Z (1,3,5)	0.520 Z (1,5)	0.412 Z (1,2,3,5)	0.594 Z (-)
ATM	0.520 Z (5)	0.372 Z (1,3,5)	0.476 Z (1,5)	0.358 Z (1,2,3,5)	0.651 Z (-)
ITM	0.527 Z (5)	0.422 Z (1,3,5)	0.461 Z (1,5)	0.405 Z (1,3,5)	0.655 Z (-)

* means significantly higher at the 10% rather than the otherwise reported 5% level.

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the contemporaneous (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; *Ad-hoc* BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. The statistical performance for the pricing errors is assessed by analyzing the proportion of theoretical prices lying outside their bid-ask spread boundaries; that is, $\text{Prob} \{ \text{Price}_{\text{model}} \notin (\text{Bid}, \text{Ask}) \}$. In each case we report the corresponding proportion for a given model and, in parentheses, the identifying number of the competing model that has a significantly higher proportion than the model being analyzed according to our Z-statistics at the 5% level. OTM, ATM, ITM are out-of-the-money, at-the-money, in-the-money options respectively. X is the exercise price, and SP is the relative bid-ask spread.

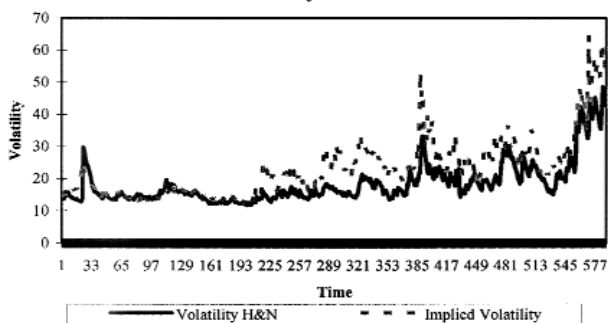
for the best models, while the frequencies under BS and HN go up by approximately 14%. Of course, these models present very high proportions outside the spread for both the in-sample and out-of-sample cases. It seems therefore that daily market conditions change considerably either because of variations of moments in the underlying distribution of returns or because demand conditions vary sufficiently to move prices quite a lot given the limited supply of contracts provided by market makers. It is probably a combination of the two factors that plays the key role in explaining out-of-sample performance. Both rapidly changing market conditions which affect the distribution of returns of the underlying asset and a limited supply of options having a strong impact on hedging (and trading) costs lead towards instability of parameters in option pricing models. Thus, our traditional theoretical framework fails in the out-of-sample context. Along this line of reasoning, it should be pointed out that recent papers using Spanish data and variants of an approximation of the risk-neutral density of terminal underlying prices by the lognormal Gram-Charlier series expansion also obtain poor out-of-sample performance. Serna (2004) employs this method to implicitly estimate the risk-neutral skewness and kurtosis of the underlying asset. Although a more consistent performance than BS is reported, pricing errors remain quite substantial. Prado (2004) argues that under this expansion one may obtain a negative risk-neutral density for some values of skewness and excess kurtosis. He incorporates the adjustment suggested by Jondeau and Rockinger (2001) to guarantee the positivity of the Gram-Charlier risk-neutral density and, as before, out-of-sample performance is quite disappointing. Similar results are obtained by León, Mencía and Sentana (2004) using semi-nonparametric densities of Gallant and Nychka (1987) which are always positive and more general than the truncated Gram-Charlier expansions. Hence, relaxing the assumption of log-normality does not seem to be sufficient to adequately price options out-of-sample.

It is also the case that liquidity costs, as proxied by the past bid-ask spreads, do not improve option pricing performance. This is consistent with the evidence reported by Peña, Rubio and Serna (2001) using a parametric approach within an out-of-sample context, and it suggests that past spreads are not useful in characterizing current market conditions. Again, it is difficult to rationalize this evidence without recurring to limited supply arguments in the option market.

To conclude, models based on smooth functions of volatility do not capture the underlying behavior of the underlying asset or of its volatility, and they seem to be useless in explaining the idiosyncratic characteristics of the internal organization of option markets.

The extremely poor performance of the HN model is also striking but, at the same time, very informative on what is missing from GARCH option pricing. Figure 1 contains the daily implied BS volatility and the daily conditional volatility estimated by expression [10]. During most of the sample period, the volatility estimated by the HN GARCH model undervalues the implied volatility from the cross-section of available options. This is a very important point. The implicit estimation of the skewness parameter is not sufficient to introduce the information contained in the cross-section of option prices¹⁷. It should be noted that the HN model is the only one in which the estimate of volatility does not employ option data. In our case, this makes its pricing performance to be very poor relative to all other models, including the traditional BS method¹⁸. Given the lack of integration between the underlying and option markets, employing parameters estimated directly from the underlying return process does not seem to help out-of-sample pricing¹⁹.

FIGURE 1
Daily annualized volatilities: Heston and Nandis versus implied volatility 1996-1998



Independently of the above arguments, the GARCH methodology is not clearly appropriate for inferring future volatility in the option pricing context. It should be recalled that we are dealing with short-term

¹⁷Despite the fact that Christoffersen and Jacobs (2003) argue that GARCH models with volatility clustering and standard asymmetric effects like the one suggested by HN perform well compared to less parsimonious alternative models.

¹⁸Similar results are found when the sample is divided into three independent years.

¹⁹Not even in-sample performance.

option data. In fact, HN report that their model does not improve on the *ad-hoc* BS model when they price the shortest to maturity options in their US database. Their GARCH modeling of variance is not able to reproduce the rapidly increasing behavior of conditional volatility needed to price short-term options, despite the fact that the model adapts to changes in volatility associated with changes in market levels. Note that jumps in returns can generate large movements, but the impact of a jump is transitory and consequently does not affect future prices. On the other hand, conditional volatility is persistent but it may only increase via small gradual normally distributed steps. In order to allow conditional volatility of returns to increase quickly, it becomes necessary to permit jumps in volatility on the underlying data generating process of equity returns. This is the point raised by Eraker, Johannes and Polson (2003) and Eraker (2004) who model jumps in volatility with constant arrival intensity and constant amplitude. However, as mentioned in our introductory comments, this approach is not flexible enough to explain the cross-section of option prices. Moreover, GARCH option modeling with jumps in volatility has not yet been developed, even under the simplest specification.

Finally, it has been recently argued by Ghysels, Santa-Clara and Valkanov (2004) that the so called mixed data sampling regression (MIDAS) framework is more appropriate than GARCH modeling when predicting volatility. Its success is based on the additional power of using more data, estimating less parameters and, especially, because of the more flexible (and optimum) weighting scheme employed when incorporating past volatility. In particular, the asymmetry in the response of the conditional volatility to positive and negative returns is more complex than previously recognized by the GARCH family. According to their evidence, negative shocks have a higher immediate effect but are ultimately dominated by positive innovations. Simultaneously, there is a strong asymmetry in the persistence of positive and negative shocks, with positive shocks being responsible for the persistence of the conditional volatility for horizons beyond two or three weeks of trading²⁰. By construction, these effects are not captured by any GARCH model. This recent evidence casts doubts on the validity of GARCH option modeling and it may also be responsible in part for the extremely poor performance of the HN model.

²⁰These results have been confirmed by León, Nave and Rubio (2004) using European equity indices including the IBEX-35 index.

The out-of-sample performance of pricing errors using the APE statistic tends to give similar results. They are reported in Table 6. The *ad-hoc* BS pricing model and the flexible univariate semiparametric option model are the best performing models. As before, there is a slight advantage for the *ad-hoc* BS, but the differences between the two models are less pronounced than when using proportions lying outside the bid-ask spread.

TABLE 6
Daily out-of-sample absolute mean pricing errors of alternative option pricing models for calls and puts

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	7.332 χ^2 (5)	5.598 χ^2 (1,3,5)	6.053 χ^2 (1,5)	5.572 χ^2 (1,3,5)	10.476 χ^2 (-)
OTM	7.920 χ^2 (5)	5.602 χ^2 (1,3,4*,5)	5.995 χ^2 (1,5)	5.691 χ^2 (1,3,5)	11.318 χ^2 (-)
ATM	5.981 χ^2 (5)	5.471 χ^2 (1,3,5)	5.914 χ^2 (5)	5.242 χ^2 (1,2,3,5)	9.396 χ^2 (-)
ITM	8.135 χ^2 (5)	5.985 χ^2 (1,3,5)	6.844 χ^2 (1,5)	5.916 χ^2 (1,3,5)	8.889 χ^2 (-)
1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	7.852 χ^2 (5)	5.724 χ^2 (1,3,5)	7.247 χ^2 (1,5)	5.688 χ^2 (1,3,5)	9.299 χ^2 (-)
OTM	7.699 χ^2 (5)	5.530 χ^2 (1,3,5)	7.164 χ^2 (1,5)	5.396 χ^2 (1,2,3,5)	8.531 χ^2 (-)
ATM	7.328 χ^2 (5)	5.329 χ^2 (1,3,5)	6.625 χ^2 (1,5)	5.319 χ^2 (1,3,5)	9.888 χ^2 (-)
ITM	11.395 χ^2 (5)	9.136 χ^2 (1,4,5)	9.326 χ^2 (1,4*,5)	9.927 χ^2 (1,5)	14.201 χ^2 (-)

* means significantly higher at the 10% rather than the otherwise reported 5% level.

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the contemporaneous (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; *Ad-hoc* BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. The pricing error is computed as the square root of the average of the squared difference between the theoretical price and the market price for each model:

$$APE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{i,model} - P_{i,market})^2}$$

The statistical performance for the pricing errors is assessed by a GMM overidentifying test with the Newey-West weighting covariance matrix. The test statistic is distributed as a χ^2 with one degree of freedom. In each case we report the corresponding absolute value pricing error for a given model and, in parentheses, the identifying number of the competing model that has a significantly different (higher) pricing error than the model being analyzed according to our χ^2 -statistics at the 5% level. OTM, ATM, ITM are out-of-the-money, at-the-money, in-the-money options respectively. X is the exercise price, and SP is the relative bid-ask spread.

5. Hedging performance

The analysis of hedging performance follows Bakshi, Cao and Chen (1997), in which a single instrument is employed. The objective is to hedge a short position for a call option with τ periods to expiration and exercise price X . Let Δ_F be the number of shares in the underlying asset to be purchased and let $B_0 = c - \Delta_F F$ be the residual position, so that the value of a replicating portfolio at t is $B_0 + \Delta_F F(t)$. Solving the standard minimum variance hedging problem, the option delta, Δ_F , has the expression

$$\Delta_F = \frac{\partial c(F, X, r, \tau, \sigma)}{\partial F} = e^{-r\tau} P_1 = e^{-r\tau} N(d_1) \quad [17]$$

where we recall that the term d_1 is given in equation [1].

Therefore, the different methods proposed in Section 2 for estimating volatility lead to different deltas. If the HN formula is used, the delta of the option has also the form $\Delta_F = e^{-r\tau} P_1$, where P_1 is different from the normal distribution, and it depends on the parameters in the GARCH model, as described in Appendix A.2.

For all models, we assume that portfolio rebalancing takes place at intervals of length δt (either a day or a week)²¹. Once the delta is estimated for each option in the sample, we obtain the resulting cash position as $B_0 = c - \hat{\Delta}_F F$ which we invest in the equivalent maturity risk free bond. At time $t + \delta t$ we calculate the hedging error for each model as

$$H(t + \delta t) = \hat{\Delta}_F F(t + \delta t) + B_0 e^{r\delta t} - c(t + \delta t, \tau - \delta t) \quad [18]$$

For computation of the hedging errors, we employ the first option in each class (same time to maturity and exercise price) that appears in the 45-minute window between 16:00 and 16:45 for both t and $t + \delta t$ ²².

To analyze the differences in hedging behavior between our competing models, we calculate the average absolute hedging error for each model as

$$H = \frac{1}{n} \sum_{i=1}^n |H_i| \quad [19]$$

²¹ Although the hedging errors become larger when one week is employed, qualitative conclusions are the same, and the results reported below only contain portfolio rebalancing taking place at intervals of one day.

²² As an alternative, the hedging performance for the last option traded in each class during each day has been analyzed and the results remain the same.

where n is the number of options over the complete sample period.

To test statistically whether the mean errors of two competing models are significantly different from zero, we again perform the GMM overidentifying restriction test, with the Newey-West weighting covariance matrix, given in equation [16]. We now have the following set of moment conditions for hedging errors and for any two competing models,

$$\begin{pmatrix} n^{-1} \sum_{i=1}^n |H_i^1| - m \\ n^{-1} \sum_{i=1}^n |H_i^2| - m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad [20]$$

where $H_i^{1,2}$ is the hedging error for option i and either model 1 or 2, and m is the common mean hedging error under the null hypothesis that both models have the same hedging error. As in our previous test, this statistic is distributed as a χ^2 with one degree of freedom.

Table 7 contains the daily hedging results. Slightly less clear conclusions may be drawn in regard to pricing performance. For puts, both the semiparametric NP(X) model and the *ad hoc* BS have lower mean hedging errors. However, for calls, both semiparametric models are superior to the *ad hoc* BS. In fact, the model with liquidity costs has the lowest mean hedging error for ATM calls. Finally, on pricing performance the HN method provides the worst results, with a surprisingly high hedging error size. Apart from the problems of historical estimation of conditional variance, another reason behind its poor behavior may be the high variability of the daily estimate of the skewness parameter. This suggests that, although this parameter is clearly important in the pricing formula, the daily variation is so high that it disturbs the estimation one day ahead rather than improving it.

In any case, as before, the overall results are quite disappointing. High performance and hedging errors are found independently of the model analyzed. The next section investigates possible biases behind these results.

TABLE 7
Daily hedging errors of alternative option pricing models for calls
and puts

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	11.694 χ^2 (5)	10.038 χ^2 (1,3,4,5)	10.170 χ^2 (1,5)	10.122 χ^2 (1,3,5)	24.549 χ^2 (-)
OTM	14.101 χ^2 (5)	11.516 χ^2 (1,3,4*,5)	11.816 χ^2 (1,5)	11.571 χ^2 (1,3,5)	26.792 χ^2 (-)
ATM	8.300 χ^2 (5)	8.013 χ^2 (4*,5)	7.951 χ^2 (1,2,4,5)	8.089 χ^2 (5)	24.333 χ^2 (-)
ITM	7.302 χ^2 (4,5)	7.123 χ^2 (1,4*,5)	6.800 χ^2 (1,4,5)	7.440 χ^2 (-)	9.846 χ^2 (-)
1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	<i>Ad-hoc</i> BS (4)	HN (5)
ALL	10.430 χ^2 (3,5)	8.512 χ^2 (1,3,5)	10.635 χ^2 (5)	8.757 χ^2 (1,3,5)	40.387 χ^2 (-)
OTM	10.587 χ^2 (3,5)	8.766 χ^2 (1,3,5)	11.105 χ^2 (5)	9.219 χ^2 (1,3,5)	35.032 χ^2 (-)
ATM	9.856 χ^2 (5)	7.317 χ^2 (1,3,5)	9.264 χ^2 (5)	7.564 χ^2 (1,3,5)	37.471 χ^2 (-)
ITM	11.030 χ^2 (5)	10.378 χ^2 (5)	11.290 χ^2 (5)	8.884 χ^2 (1,5)	44.511 χ^2 (-)

* means significantly higher at the 10% rather than the otherwise reported 5% level.

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the previously (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; *Ad-hoc* BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. The statistical performance of the hedging errors is assessed by a GMM overidentifying test with the Newey-West weighting covariance matrix. The test statistic is distributed as a χ^2 with one degree of freedom. In each case we report the corresponding mean hedging error for a given model and, in parentheses, the identifying number of the competing model that has a significantly different (higher) mean hedging error than the model being analyzed according to our χ^2 -statistics at the 5% level. OTM, ATM, ITM are out-of-the-money, at-the-money, in-the-money options respectively. *X* is the exercise price, and *SP* is the relative bid-ask spread.

6. The structure of pricing errors

A further analysis trying to understand the structure of the out-of-sample pricing errors of these models would seem to be called for. We now use a simple regression framework to study the relationship

between percentage pricing errors and factors that are either option-specific or market dependent. We first take as given an option pricing model, and let e_{it} be the i th option's percentage pricing error on day t defined as the theoretical price minus the market price, divided by the market price. Finally, we run the following regression for the whole sample period and for calls and puts separately:

$$e_{it} = a + b_1\tau_{it} + b_2\sigma_t + b_3SK_t + b_4KU_t + b_5 \ln X_{it} + b_6SP_{it} + \varepsilon_{it} \quad [21]$$

where τ_{it} is the annualized time to maturity of the i th option at day t , σ_t is the annualized daily standard deviation of the IBEX-35 index returns computed from 15-minute intradaily returns, and similarly SK_t and KU_t are the daily skewness and the (excess) kurtosis at day t respectively; X_{it} is the exercise price and SP_{it} is the relative bid-ask spread of the i th option. The results are reported in Table 8, where the results for calls and puts are reported separately. As can be observed, the explanatory variables are significantly different from zero in almost all cases. These results provide evidence against models trying to explain the time-varying behavior of the volatility function as a smooth function of previously defined variables, be it in an *ad hoc* way, in a GARCH parametric framework or in our nonparametric context. The results are consistent with the need to simultaneously incorporate a more complex behavior in the process of returns and volatility with (probably) correlated jumps and the effects of organizational characteristics of the option market.

It is interesting to notice the large biases associated with the HN model. The (negative) magnitude of the b_2 coefficient related to the model is very high and significant. It is the largest coefficient (in absolute value) of all the models, and it reflects that the conditional variance estimated with historical return data undervalues the variance reflected in option prices, particularly when there is a lot of variability in the market. This is clearly consistent with Figure 1. As expected, the second largest coefficient (in absolute value) comes from the BS model, where constant volatility across options is imposed. Moreover, again for these two models, the b_1 coefficients are also large. Thus, they reflect a large time to maturity bias for both calls and puts. As reflected in the magnitude of the coefficients, the most problematic pricing performance of the two models tends to be associated with options with the longest time to expiration. Hence, the deficiencies discussed for these models become more pronounced with options that

TABLE 8
Percentage pricing errors and explanatory variables

1996-1998 CALLS	BS (1)	NP (X) (2)	NP (X, SP) (3)	Ad-hoc BS (4)	HN (5)
Intercept	-1.069 (0.000)	-0.173 (0.000)	-0.209 (0.000)	-0.049 (0.268)	-0.050 (0.680)
Time to maturity	0.562 (0.000)	-0.031 (0.586)	-0.114 (0.065)	-0.082 (0.187)	3.420 (0.000)
Volatility	-0.455 (0.000)	-0.163 (0.000)	-0.169 (0.000)	-0.115 (0.000)	-0.997 (0.000)
Skewness	-0.041 (0.000)	-0.019 (0.000)	-0.019 (0.000)	-0.015 (0.000)	-0.008 (0.000)
Kurtosis	-0.001 (0.536)	-0.000 (0.705)	0.001 (0.291)	0.003 (0.002)	0.004 (0.157)
Ln Exercise	0.124 (0.000)	0.020 (0.000)	0.024 (0.000)	0.005 (0.323)	-0.001 (0.945)
Spread	0.212 (0.000)	0.057 (0.000)	0.080 (0.000)	0.062 (0.000)	0.785 (0.000)

1996-1998 PUTS	BS (1)	NP (X) (2)	NP (X, SP) (3)	Ad-hoc BS (4)	HN (5)
Intercept	-0.279 (0.000)	-0.657 (0.000)	-0.886 (0.000)	-0.335 (0.000)	-0.597 (0.000)
Time to maturity	-0.496 (0.000)	-0.036 (0.601)	-0.294 (0.000)	0.183 (0.016)	1.737 (0.000)
Volatility	-0.200 (0.000)	-0.119 (0.000)	-0.137 (0.000)	-0.064 (0.007)	-0.619 (0.000)
Skewness	0.021 (0.000)	0.032 (0.000)	0.035 (0.000)	0.049 (0.000)	0.045 (0.000)
Kurtosis	-0.003 (0.003)	-0.004 (0.000)	-0.004 (0.005)	-0.001 (0.248)	-0.001 (0.620)
Ln Exercise	0.041 (0.000)	0.082 (0.000)	0.112 (0.000)	0.043 (0.000)	-0.065 (0.000)
Spread	-0.299 (0.000)	-0.069 (0.000)	-0.072 (0.000)	-0.014 (0.122)	0.146 (0.000)

BS is the Black-Scholes model; NP(X) is the semiparametric option model, where the volatility function is estimated nonparametrically for each day from January 2, 1996 to November 10, 1998 using all available call/put options transacted from 11:00 to 16:45, and considering only the exercise price as an explanatory variable. The call/put price is calculated by using the BS pricing function evaluated at the previously (nonparametrically) estimated volatility; NP(X, SP) is a similar price except that the nonparametric estimated volatility includes not only the exercise price but also the bid-ask spread of each option as a proxy for liquidity; Ad-hoc BS is a version of the BS model with exercise specific implied volatility; HN is the closed-form GARCH model proposed by Heston and Nandi (2000) estimated with an updated asymmetric GARCH. X is the exercise price, and SP is the relative bid-ask spread. For a given option pricing model, the following regression is employed to explain the percentage pricing errors of all call/put options transacted in our sample:

$$e_{it} = a + b_1\tau_{it} + b_2\sigma_t + b_3SK_t + b_4KU_t + b_5\ln X_{it} + b_6SP_{it} + \varepsilon_{it}$$

where for the i th option at date t , e_{it} is the percentage pricing error defined as the difference between the theoretical price and the market price divided by the market price; τ_{it} the time to maturity; X_{it} the exercise price and SP_{it} the bid-ask spread. σ_t , SK_t and KU_t are respectively the daily volatility, skewness and (excess) kurtosis. P-values in parentheses.

have more days to maturity. In any case, it should be recognized that, at least for puts, all pricing models except the univariate semiparametric case have more difficulties in pricing options as time to expiration becomes longer.

Finally, the coefficient associated with the bid-ask spread is again large in the HN case. If the underlying asset and the option market are not fully integrated, any model estimating most relevant parameters only from the stock market will be unable to capture idiosyncratic characteristics of option prices.

In short, neither model seems to reflect appropriately the underlying distribution characteristics of the IBEX-35 and/or the idiosyncratic characteristics of the option market microstructure.

7. Conclusions

In this paper, we employ intraday option data from the Spanish market to test both the pricing and hedging performance of the five option pricing models. The results show that, from the out-of-sample point of view, simplicity is an important characteristic in the option pricing framework used. The *ad-hoc* BS and the simplest univariate semiparametric models are the best performing models. In fact, the overall picture suggests that the *ad-hoc* BS is slightly superior despite the flexibility added by the nonparametric estimation of the volatility function. However, the overall out-of-sample performance of all models is quite poor. On the other hand, in-sample pricing performance shows that liquidity cost is a key issue in option pricing. If our liquidity costs reflect the net buying pressure from public orders, our evidence may indicate that there exists a strong daily changing behavior in the demand for options at different exercise prices with an upward sloping supply curve, given the quite different performance of the model with liquidity costs when we move from in-sample to out-of-sample tests.

Simultaneously, the overall poor performance of the models may also suggest that (probably) correlated jumps in returns and volatility are a key feature to be adopted for any competitive option pricing model. It seems that the volatility function is not a smooth function of the underlying variables used in the estimation. Also, and somewhat surprisingly, the HN specification presents a very poor pricing and hedging performance. The GARCH framework cannot generate (time-varying) skewness and kurtosis in the degree needed to price options. In other

words, GARCH volatility does not incorporate the rich information contained in the cross-section of option prices, in spite of the fact that the asymmetric GARCH parameter is estimated implicitly from option data. The volatility inferred from the history of the index returns is not high enough to obtain reasonable option prices.

Our results, together with the evidence currently available from stochastic models with volatility and jump risk factors, suggest that an integrated approach of both options and stock markets that also incorporates correlated jumps in volatility may be a promising area of research. Simultaneously, explicit analysis of financial intermediation of the underlying risks by option market makers and the effects of time-varying net buying pressure along with upward sloping supply curves in option prices is probably more effective. Given our experience with option data, we support microstructure explanations rather than more elaborate (and difficult to estimate) models²³. In any case, further research is clearly justified with a view to understanding the seemingly rapidly changing behavior of the underlying equity asset return distribution and net buying pressure conditions as described by Bollen and Whaley (2004).

Appendix A1. The procedure in the nonparametric estimation

We summarize the main results that have motivated the nonparametric estimation process employed in this work. Proofs and specific details can be found in Ferreira and Gago (2002).

A1.1 Kernel and SNN properties

A second order kernel is a function $K(\cdot)$ such that $\int K(u) du = 1$; $\int K(u) u du = 0$; $\int K(u) u^2 du = d_K < \infty$ and $\int K(u)^2 du = c_K < \infty$. The Gaussian kernel is used in our empirical analysis. It is defined as

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$$

²³It must be recognized that we are dealing with options characterized by very short-term to expiration traded in a rather thin option trading market. The results should be taken under this perspective. For example, Huang and Wu (2004) show that the factors dominating short-term and long-term options are substantially different when pricing options within large markets.

In the general regression problem written as, $E(Y/X = x) = m(x)$, and if an *i.i.d.* sample $\{x_i, y_i\}_{i=1}^n$ is available, the mean square errors obtained for the kernel (MSE_K) and the SNN estimators (MSE_{SNN}) are respectively given by

$$\begin{aligned} MSE_K(x) &= BIAS_K^2(x) + VAR_K(x) = \\ &= \frac{d_K^2 h^4}{4} \frac{(m''f + 2m'f')^2(x)}{f^2(x)} + \frac{\sigma^2}{nhf(x)} c_K \end{aligned}$$

$$\begin{aligned} MSE_{SNN}(x) &= BIAS_{SNN}^2(x) + VAR_{SNN}(x) = \\ &= \frac{d_K^2 h^4}{4} \frac{(m''f - m'f')^2(x)}{f^6(x)} + \frac{\sigma^2}{nh} c_K \end{aligned}$$

where the first term in the sum corresponds to the squared bias ($BIAS^2$), and the second to the variance term (VAR). Moreover, $f(x)$ is the density function of the explanatory variable X and σ^2 the variance of $Y/X = x$.

A common global error measure is the mean integrated squared error, $MISE = \int MSE(x) dx$. The minimum value for $MISE$ are attainable by selecting the optimal smoothing parameter $h = \arg \min_h MISE$.

By substituting the optimal smoothing parameter in the $MISE$, its minimum value is obtained. It can be shown that, under very general situations, the $MISE_{SNN}$ is smaller than the $MISE_K$ in the tails of the distribution; that is, in those zones where the data density is low.

In a multivariate setting, the expressions for the $MISE$ are more complicated due to the bias term. However, the variance term is easier to compute and the expressions of a d -dimensional kernel and a SNN estimator are respectively,

$$\begin{aligned} VAR_K &= \frac{\sigma^2 c_K}{nh_1 \dots nh_d} \frac{1}{f(x_1, \dots, x_d)} \\ VAR_{SNN} &= \frac{\sigma^2 c_K}{nh_1 \dots nh_d} \frac{f_1(x_1) \dots f_d(x_d)}{f(x_1, \dots, x_d)} \end{aligned}$$

Thus, in the multivariate setting, the variance term corresponding to the SNN estimator now depends on the joint density, unless the covariates are independent. However, it can be shown that for a wide class of densities, a global bandwidth in the SNN estimator still leads

to a more stable variance than the kernel estimator. In fact, the variance of the kernel estimator can increase drastically in the tails, while the variance of the SNN estimator remains bounded. This fact explains the superior behavior of the SNN estimator method even when a multivariate setting is employed.

A1.2 Smoothing parameter selection

The plug-in criteria are based on the expression for the optimal h that minimizes the *MISE*. The objective is to directly compute the value for h where the unknown quantities are substituted by their estimators. In the univariate case, only the second derivative of $m(\cdot)$ is unknown. Gasser, Kneip and Köhler (1991) propose an algorithm that uses the following basic steps: (i) compute $h_0 = 1/n$; (ii) given h_{j-1} , estimate the second derivative of $m(\cdot)$ using $h_{j-1}n^{1/10}$; (iii) compute h_j in the expression for the optimal bandwidth with the estimation of $m''(\cdot)$; (iv) stop when h_{j-1} is close to h_j . This algorithm is very easy to implement and the empirical results are satisfactory. However, when a bivariate estimator is computed, the selection procedure must take into account the presence of additional variables. Furthermore, it is desirable to apply alternative nonparametric estimators with different smoothing parameters to check the robustness of the results in the selection process.

In our case, and in order to take these ideas into account, the univariate criterion has been employed in our bivariate framework as a way to obtain pilot parameters. Then, as a multivariate criterion, the natural extensions of the Generalized Cross Validation (*GCV*) and Rice methods are employed. Both belong to the class of criteria based on the minimization of a penalized version of the residual sum of squares (*RSS*) of the form:

$$G(h) = RSS(h) \phi(n^{-1}h^{-1})$$

where

$$RSS(h) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}(x_i))^2$$

where ϕ is the penalizing function. Any of the nonparametric estimators employed can be written in a vector form as $\hat{m} = K(h)y$, where $\hat{m}$ is the vector containing all estimators at the design points, y is the vector of observations for the dependent variable, and $K(h)$ is

the projection matrix containing the proper weights for each estimator considered. For the *GCV* criterion,

$$\phi_{GCV}(n^{-1}h^{-1}) = \left(1 - \frac{\text{tr}K(h)}{n}\right)^{-2}$$

and for Rice's,

$$\phi_R(n^{-1}h^{-1}) = \left(1 - 2\frac{\text{tr}K(h)}{n}\right)^{-1}$$

All three selection methods are implemented in the empirical analysis, leading to similar estimators, but only the results based on Rice's criterion are reported in the tables.

Appendix A2. The Heston and Nandi (HN) option pricing formula

In the computations, daily returns are considered and therefore Δ is set equal to 1 in the HN formula. In this setting, the HN risk-neutral process is given by

$$\begin{aligned}\ln S(t) &= \ln S(t-1) + r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t) \\ h(t) &= \omega + \beta h(t-1) + \alpha \left(z^*(t-1) - \gamma^* \sqrt{h(t-1)}\right)^2\end{aligned}$$

where

$$\begin{aligned}z^*(t) &= z(t) + \left(\lambda + \frac{1}{2}\right) \sqrt{h(t)} \\ \gamma^* &= \gamma + \lambda + \frac{1}{2}\end{aligned}$$

The option pricing formula is then given by

$$c_{HN} = e^{-r\tau} [F(t) P_1 - X P_2]$$

where

$$\begin{aligned}P_1 &= \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[\frac{X^{-i\phi} f^*(i\phi + 1)}{i\phi f^*(1)} \right] d\phi \\ P_2 &= \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[\frac{X^{-i\phi} f^*(i\phi)}{i\phi f^*(1)} \right] d\phi\end{aligned}$$

The function $f^*(\phi)$ corresponds to the conditional generating function of the risk-neutral process of the asset price. In this case, it takes the log-linear form

$$f^*(\phi) = E_t^* \left[S(T)^\phi \right] = S(t)^\phi e^{A(t;T,\phi) + B(t;T,\phi)h(t+1)}$$

The terms $A(\cdot)$ and $B(\cdot)$ depend on the set of parameters $(\omega, \alpha, \beta, \gamma^*)$ and the estimation procedure is performed in a recursive form as

$$\begin{aligned} A(t;T,\phi) &= \\ &= A(t+1;T,\phi) + B(t+1;T,\phi)\omega - \frac{1}{2} \ln [1 - 2\alpha B(t+1;T,\phi)] \\ B(t;T,\phi) &= \\ &= \phi \left(\gamma^* - \frac{1}{2} \right) - \frac{1}{2} \gamma^{*2} + \beta B(t+1;T,\phi) + \frac{1/2 (\phi - \gamma^*)^2}{1 - 2\alpha B(t+1;T,\phi)} \end{aligned}$$

with the initial conditions $A(T;T,\phi) = B(T;T,\phi) = 0$

As described in the text, all parameters are computed from the GARCH specification and they are used to estimate $h(t+1)$. The options pricing formula is then used to estimate the implicit γ^* as

$$\gamma^* = \arg \min_{\gamma} \sum_{i \in t} (c_{HN,i}(\gamma) - c_i)^2$$

where c_i is the price observed in the market, and $c_{HN,i}(\gamma)$ denotes the price resulting from the HN expression.

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Resumen

Este trabajo presenta una comparación de modelos alternativos de valoración de opciones que no utilizan procesos generadores de datos basados en difusión con saltos o volatilidad estocástica. Suponemos una función de volatilidad suave de variables explicativas definidas previamente o un modelo en el cual, usando directamente observaciones discretas, podemos estimar volatilidades dependientes y una correlación negativa entre volatilidad y los rendimientos subyacentes. También admitimos fricciones de liquidez para reconocer que los mercados subyacentes pudieran no estar integrados. Los modelos más sencillos tienden a presentar resultados superiores fuera de muestra y una mejor capacidad de cobertura, aunque el modelo con costes de liquidez parece tener un mejor comportamiento dentro de la muestra. Sin embargo, ninguno de los modelos es capaz de capturar los rápidos cambios en la distribución de los rendimientos del índice subyacente, ni la presión neta de compra que caracteriza a los mercados de opciones.

Palabras clave: Valoración de opciones, volatilidad condicional, cobertura, liquidez, presión neta de compra.

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